

Medical diseases complicating pregnancy

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LEARNING OBJECTIVES

- Understand the importance of medical conditions in pregnancy in relation to maternal and infant health.
- Appreciate the importance of preconceptual counselling and its impact on improving pregnancy outcomes.
- Understand the impact of common medical conditions such as hypertension, kidney disease, cardiac disease and diabetes on pregnancy.
- Appreciate the contribution of maternal medical disease to maternal mortality

overview

- Pregnancy is a normal physiological event during which most women remain fit and well.
- However, it may cause specific medical conditions.
- It is important that women receive good advice pre-pregnancy about the potential impact of their medical condition and enter pregnancy with appropriate confidence about **routine medication or specific management plans to alter treatment in the first trimester**



What the pregnant woman wants to know ?

- **Will I have a normal healthy baby?**
- **Will pregnancy make my disease worse?**
- **Will my disease be worse after pregnancy?**
- **Is there a risk of my children inheriting my condition?**
- **What treatment is safe during pregnancy?**
- **Can I have an epidural?**
- **Should I be delivered by caesarean section?**
- **Is breastfeeding advisable?**

Endocrine disease

- **Thyroid disease** is the commonest endocrine disorder in pregnant women and this will therefore be considered in more detail than other endocrinopathies.

Thyroid disease:

Thyroid disease is common in women of child bearing age.

- **However, symptoms of thyroid disease:**

such as **heat intolerance, constipation, fatigue, palpitations** and **weight gain** resemble those of normal pregnancy

Physiological changes

- During normal pregnancy the **circulating levels of thyroid binding globulin increase**, and as a consequence total T3 and T4 levels also increase.
- **Therefore the free hormone levels should be measured in pregnant women.**
- TSH levels should be interpreted with caution in the first trimester as **hCG has a weak stimulatory effect on the TSH receptor.**

- **The fetus cannot synthesize thyroxine until the 10th week of gestation**, and it is therefore dependent upon transplacental transfer of maternal hormone.

There is increased maternal synthesis of thyroid hormones in the first trimester as a **result of transplacental passage and the high levels of thyroid binding globulin.**

Physiological change ..cont.

- this in turn results in an increased maternal requirement for iodide.
- In areas of **relative iodide deficiency** this may result in the development of **maternal hypothyroxinaemia and goitre.**

HYPOTHYROIDISM

- Hypothyroidism affects approximately **1%** of pregnant women.
- Providing **thyroxine replacement therapy is adequate**, hypothyroidism is not associated with an adverse pregnancy outcome for the mother or fetus.

poorly controlled hypothyroidism and a including :variety of adverse outcomes

1. subfertility.
2. congenital abnormalities,
3. **hypertension,**
4. increased risk of miscarriage
5. **premature delivery,**
- 6.fetal growth restriction and
7. **post-partum haemorrhage.**
- 8.placental abruption .

Hypothyroidism

- **Severe hypothyroidism affects the subsequent intelligence of the offspring of affected mothers.**

Women with hypothyroidism should be given thyroxine replacement at a dose that ensures their thyroid function tests are normal with a **FT4** at the upper end of the normal range appropriate for each trimester of pregnancy.

- Thyroxine absorption is decreased by certain drugs including **iron and calcium supplements**.
- thyroxine is best taken on an empty stomach and 4 h apart from any iron or other supplements.

HYPERTHYROIDISM:

- Hyperthyroidism affects **1 in 500** pregnant women, **90%** of whom have **Graves' disease**.
- **Graves' disease is caused by TSH receptor stimulating antibodies.**
- Women with well-treated disease rarely have maternal complications of pregnancy
- the disease may remit during the latter trimesters such that **treatment may need to be reduced or stopped.**

- In the post-partum period the disease may flare and require treatment with the same or **higher doses of antithyroid medication.**

Poorly controlled hyperthyroidism is associated with pregnancy complications

- Including : **1.maternal thyrotoxic crisis,
2.miscarriage,
3. gestational hypertension,
4. pre-eclampsia and
5. intrauterine growth restriction .**
- **The risk of these complications is reduced if the disease is adequately controlled before delivery.**

Treatment of hyperthyroidism

The principal drugs used is (**propylthiouracil, carbimazole**) inhibit thyroid hormone synthesis.

- a **greater proportion of carbimazole enters breast milk**, and therefore **propylthiouracil** is usually the drug of choice if a woman is diagnosed as having hyperthyroidism for first time during pregnancy

- Both drugs may rarely cause **neutropenia and agranulocytosis**.
- Therefore patients should be aware that symptoms of **infection**, particularly **sore throat**, may be associated with bone marrow suppression and they must have a **neutrophil count checked**

- Once drug treatment has been commenced **thyroid function tests should be carried out and checked regularly.**

Propylthiouracil and carbimazole both cross the placenta, fetal hypothyroidism is rarely seen.

TSH receptor stimulating antibodies also cross the placenta and may influence the fetal and neonatal thyroid status.

Heart Disease in Pregnancy

- **Normal physiological changes of pregnancy that mimic symptoms and signs of cardiac disease**
- **Symptoms**
 - Tiredness
 - Dyspnoea
 - Orthopnoea
 - Syncope

- **Electrocardiogram**

- Left axis deviation
- ST segment and T wave changes
- Small Q, inverted P or T wave in lead III
- Increased R wave amplitude in lead V2
- Atrial or ventricular ectopics

- **Chest X-ray**

- Straightened left upper cardiac border
- Horizontal heart position
- Increased lung markings

Functional classification

- According to New York Heart Association (1964);
- Class I: No discomfort (i.e. dyspnoea, palpitation or anginal pain) on ordinary activity.
- Class II: Discomfort on ordinary activity.
- Class III: Discomfort on less than ordinary activity.
- Class IV: Dyspnoea at rest. Patient is decompensated.

- **1. Low risk of maternal mortality (less than 1%).**
 - (a) Septal defects.
 - (b) New York Heart Association classes I and II.
 - (c) Patent ductus arteriosus.
 - (d) Pulmonary / tricuspid lesions.
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- **2. Moderate risk of maternal mortality (5-15%).**
 - (a) NYHA classes III and IV mitral stenosis.
 - (b) Aortic stenosis.
 - (c) Marfan's syndrome with normal aorta.
 - (d) Uncomplicated coarctation of aorta.
 - (e) Past history of myocardial infarction

- **3. High risk of maternal mortality (25-50%).**
 - (a) Eisenmenger's syndrome.
 - (b) Pulmonary hypertension.
 - (c) Marfan's syndrome with abnormal aortic root.
 - (d) Peripartum cardiomyopathy.

Effect of Pregnancy on Heart Disease

- **Heart failure:**

- During pregnancy, heart failure can occur at any time but the maximum incidence is between 32 and 34 weeks when the blood volume and cardiac output are in their peaks.
- After that they have a plateau level up to full term.
- During the 2nd stage, heart failure may occur due to stress on the heart.

- After delivery, failure may occur due to loading of the circulation by the blood from the placental sinuses after retraction of the uterus.
- Subacute bacterial endocarditis: may develop in the puerperium.



Risk factors for the development of heart failure in pregnancy

- **Respiratory or urinary infections**
- **. Anaemia.**
- **Obesity.**
- **Corticosteroids.**
- **Tocolytics.**
- **Multiple gestation.**
- **Hypertension.**
- **Arrhythmias.**
- **Pain-related stress.**
- **Fluid overload.**

Effect of Heart Disease on Pregnancy

- Abortion.
- Intrauterine growth retardation.
- Still birth.
- Premature labour.
- These complications are encountered especially in cyanotic heart diseases.



Management

Pre-conceptual counselling

- This is an important aspect of management for the cardiac patient planning a pregnancy.
- Ideally, the obstetrician and cardiologist should work together to help the patient make an informed decision.
- Prevent an unwanted pregnancy and avoid the risks associated with pregnancy continuation or termination.



Toronto risk markers for maternal cardiac events

- 1-prior episode of heart failure, arrhythmia or stroke
- 2-NYHA class $\geq$ II or cyanotic
- 3- left heart obstruction (**mitral valve area $< 2.0 \text{ cm}^2$, aortic valve area $< 1.5 \text{ cm}^2$, or peak left ventricular outflow tract gradient $> 30 \text{ mm Hg}$**)
- 4-reduced left ventricular function ($\text{EF} < 40\%$)



General management:

- More frequent antenatal visits.
- More rest.
- Diet is directed to restrict weight gain and prevent anaemia as it increases cardiac strain.
- Infection should be avoided and properly treated.
- Hospitalisation: if signs of decompensation occur, the earliest evidence is tachycardia exceeding 100 beats/ minute and crepitations at the lung bases. Rest in a hospital is desirable in special conditions.



- Beta-adrenergic blockers: as propranolol may be indicated for arrhythmia associated with ischaemic heart disease.
- Aminophylline: relieves bronchospasm.
- Heparin: is indicated in patients with artificial valves or atrial fibrillation.

- N.B. Acute pulmonary oedema is urgently treated by:
 - Morphine 15 mg IV, to allay anxiety and reduce venous return.
 - Oxygen.
 - Digoxin 1 mg IV, except in severe mitral stenosis as the increase in right heart output cannot be handled by the mitral valve.
 - Aminophylline 250 mg IV.
 - Venesection, removing 500 ml blood rapidly may be indicated in severe cases.

- **Cardiac surgery:**

- It may be an alternative to therapeutic abortion. The principal indication is recurrent pulmonary oedema with mitral stenosis and heart failure not responding to medical treatment.
- There is no increased risk to the mother or the foetus in closed cardiac surgery e.g. mitral valvotomy but there is higher incidence of foetal loss with open surgery

Management of labour in women with heart disease

- Avoid induction of labour if possible.
- Use prophylactic antibiotics essential to guard against subacute bacterial endocarditis.
- Ensure fluid balance.
- Avoid the supine position.
- Discuss regional/epidural anaesthesia/analgesia with senior anaesthetist.
- Keep the second stage short.
- Use Syntocinon judiciously.

- Shorten the second stage by forceps or ventouse.
- Ergometrine is better avoided as it causes sudden load of the circulation with blood from the uterus leading to acute heart failure.
- Postpartum observation for 48 hours is essential as the risk of heart failure is high in this period.
- Although bed rest is essential, early ambulation is desirable to avoid thromboembolism.



Contraception

- Barrier methods – unreliable.
- COC contraindicated.
- Progesterone only pill have better side effect profile & long acting slow releasing as Mirena intrauterine system have improved efficacy.
- Sterilization where family completed. (Laparoscopic clip sterilization carries risk).

Gastroenterology

- Problems such as **appendicitis**, **pancreatitis** and **cholecystitis** can arise in pregnancy.
- They must be managed aggressively to minimize any risk of associated **peritonitis** which can result in **premature labour and associated sepsis**.
- Diagnosis of such complications can be difficult and requires an experienced opinion.

- It is generally recommended that **early recourse to surgery for an acute appendicitis** is the best option to prevent the development of peritonitis with possible serious sequelae, including **preterm delivery.**

Coeliac disease

- Coeliac disease is a gluten-sensitive enteropathy with a prevalence of around 0.3–1% in the general population.
- It has been estimated that up to 1 in 70 pregnant women are affected by coeliac disease, but that many are undiagnosed.
- Untreated coeliac disease is associated with high rates of spontaneous miscarriage and other adverse pregnancy outcomes such as FGR.
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- Once a gluten-free diet is established, women with coeliac disease can expect a healthy pregnancy outcome.
- Awareness that they are predisposed to other autoimmune diseases is the only other consideration.

Inflammatory bowel disease

- Inflammatory bowel disease can also complicate pregnancy.
- Pregnancy outcome is in general satisfactory, though there may be some increased risk of **preterm birth and IUGR**.
- treated in the same way in pregnancy as in the non-pregnant state, with **steroids and sulphasalazine** the mainstays of treatment.

- The majority of patients with inflammatory bowel disease (IBD), which includes ulcerative colitis and Crohn's disease, are diagnosed during their reproductive years (incidence 3 per 1,000 pregnancies).
- Women with IBD have reduced fertility rates compared with the general population.
- Pregnancy does not usually alter the course of IBD, but new-onset IBD, uncontrolled disease activity at conception and previous bowel resection increase the risk of an adverse pregnancy outcome.

- Similarly, disease flare during pregnancy is more likely if the disease is active at the time of conception.
- Pregnant women with either ulcerative colitis or Crohn's disease have been shown to have similar rates of disease exacerbation, although flares in the first trimester and postpartum are more common in disease.
- Supplementation with high-dose folic acid (5 mg daily) is recommended and supplementation with other vitamins is indicated according to measured deficiencies.

- Women with IBD have an increased rate of delivery by caesarean section, preterm labour and small for gestational age offspring.
- or.
- Caesarean section is indicated for the usual obstetric indications, and also if there is active perianal disease in Crohn's and usually if there is an ileoanal pouch.
- The use of medication during conception and pregnancy is a cause of concern for many patients with IBD.

- Methotrexate is contraindicated, but the majority of other medications used to induce or maintain remission, including 5- aminosalicylates, azothiapriner, ciclosporin and corticosteroids, are low risk.

Liver disorders

- **Liver disorders frequently complicate pregnancy,.**
- **but fortunately rarely result in long-term morbidity.**

Liver disorders:includes

- 1.Cholestasis of pregnancy.
- 2. Acute fatty liver with pregnancy.
- 3. Hyperemesis.
- 4.Gastroenterology.
- 5.Inflammatory bowel disease.
- 6. Viral hepatitis.

Viral hepatitis:

- **Clinical significance**

The principal forms of hepatitis that complicate pregnancy are hepatitis **A, B, C, D and E.**

Hepatitis is a relatively benign clinical disorder that does not pose a serious risk



Hepatitis A

is the second most common cause of hepatitis , but it is relatively uncommon in pregnancy.

It is caused by an RNA virus that is transmitted by fecal-oral contact.

-Infections in children are usually asymptomatic; infections in adults are usually symptomatic.

Infection does not result in a chronic carrier state, and perinatal transmission essentially never occur.

Hepatitis B

Is the most common form of viral hepatitis in obstetric patients.

It is caused by a DNA virus that is transmitted parenterally and via sexual contact.

Acute hepatitis B occurs in approximately **1 to 2 per 1000 pregnancies** .

The chronic carrier stage is more frequent

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Hepatitis C

Hepatitis C is caused by an RNA virus that is transmitted parenterally, via sexual contact, blood product and perinatally.

Hepatitis D is an RNA virus that depends upon co-infection with hepatitis B for replication



Hepatitis E

Hepatitis E is caused by an RNA virus.

The epidemiology of hepatitis E is similar to that of hepatitis A.

It is endemic in developing countries of the world.,
maternal infection with hepatitis E often has an
alarmingly high mortality,.

**A chronic carrier state does not exist, and
perinatal transmission is extremely unlikely**



Clinical manifestation of Hepatitis

- The typical clinical manifestations include:
- low-grade fever,
- malaise,
- poor appetite.
- right upper quadrant pain and tenderness,
- jaundice



Diagnosis of viral hepatitis

The best test to confirm the diagnosis of **acute hepatitis A** is identification of **anti-hepatitis A-IgM antibody**.

serum concentration of direct ,indirect bilirubin



Acutely infected patients also may have
**elevated liver transaminase enzymes and
elevated**

TABLE 1 Summary of key features of viral hepatitis in pregnancy

Infection	Nucleic acid	Best diagnostic test	Perinatal transmission	Immunoprophylaxis
A	RNA	Antibody detection	Rare	Hepatitis A vaccine
B	DNA	Detection of surface antigen PCR	Common in absence of immunoprophylaxis	HBIG and hepatitis B vaccine
C	RNA	Antibody detection PCR	Uncommon unless patient is coinfectd with HIV	None
D	RNA	Antigen and antibody detection	May occur in association with transmission of hepatitis B	Prophylaxis for hepatitis B protects against hepatitis D
E	RNA	Antibody detection PCR	Rare	Hepatitis E vaccine
G	RNA	Antibody detection	Yes, but no clinical significance	None

Abbreviations: PCR, polymerase chain reaction; HBIG, hepatitis B immune globulin.

Diagnosis:

Hepatitis B virus:

has three distinct antigens:

the surface antigen (HBsAg) which is found in serum,

the core antigen (HBcAg) which is found only in hepatocytes,

the e antigen (HBeAg) which also is found in serum.

Detection of the latter antigen is indicative of an extremely high rate of viral replication

Treatment Of viral hepatitis

- Patients with acute hepatitis A require supportive therapy.
- Their nutrition should be optimized.
- Of great importance, household contacts should be **vaccinated with hepatitis A vaccine.**
- **Patients with acute hepatitis B** require similar supportive care.

Their household contacts and sexual partners should receive hepatitis B immune globulin, followed by the hepatitis B vaccine series.

- **Infants delivered to mothers with hepatitis B infection should immediately receive the hepatitis B immune globulin and first dose of hepatitis B vaccine while still in the hospita**

Complications of Viral Hepatitis

- **Obstetric complications:**
 - 1. First trimester spontaneous miscarriage.
 - 2. preterm delivery.
 - 3. Fetal growth restriction.
 - 4. stillbirth.
 - 5. Preterm rupture of membranes.
 - 6. Low birth weight and neonatal unit admission.

Pregnant women who are infected with hepatitis B pose a significant risk of transmission to their offspring

Hepatitis B, may result in chronic liver disease such as chronic active hepatitis, chronic persistent hepatitis and cirrhosis.

Chronic disease also predisposes to the development of hepatocellular carcinoma.

Most neonates become infected at time of delivery as a exposure to contaminated blood and genital tract secretions.

- Patients who are seropositive for surface antigen alone have at least a 20% risk of transmitting infection to neonate.
- Women who are seropositive for both the **surface antigen and e antigen have almost a 90% risk of perinatal transmission**

Cholestasis of pregnancy:

- -Cholestasis of pregnancy is the most common liver condition affecting pregnancy and it classically presents with an itch and consequent lack of sleep in the **third trimester**.
- -it is associated with an increased risk of intrauterine death, classically from 37 weeks' gestation.

- meconium passage and preterm labour.
- Laboratory investigations include: - liver function tests and assay of serum bile acids.
- It is currently uncertain whether the bile acids themselves may be directly responsible for fetal demise.
- _

- Treatment strategies include :
 - 1.timely delivery,
 - 2. cool aqueous menthol cream to relieve itch,
 - 3. ursodeoxycholic acid and vitamin K.
- Ursodeoxycholic acid is currently the mainstay of treatment .
- This condition has a high likelihood of recurrence (approximately 80%).

Respiratory disease:

- Women with respiratory disorders require careful assessment when they present for antenatal care
- .
- -For those with risk of respiratory compromise during pregnancy or delivery investigation with pulmonary functions tests may be necessary or exclusion of associated pulmonary vascular disease by echocardiography.

- -An anaesthetic opinion prior to the third trimester is valuable, including for those with possible respiratory compromise due to musculoskeletal problems.
- -Breathlessness can be one of the most difficult symptoms to interpret in pregnancy.

- Some increase in breathlessness arises during the course of a normal pregnancy.
- Breathlessness can be a manifestation of
 - 1.thromboembolism.
 - 2.cardiac disease .
 - 3. deterioration of background respiratory disease.
- Patients should have: -a careful clinical assessment by history and examination.
- -Oxygen saturation, -arterial blood gasses may all help in differentiating physiological

Asthma:

- Asthma is the most common respiratory disorder affecting 3% of women of childbearing age. Pregnancy has a variable effect on asthma :
- 1. 1/3 improve. 2. 1/3 deteriorate. 3. 1/3 remain the same.
- -The most common reason that their asthma symptoms deteriorate is patients reducing their treatment because of a belief that the medication may be harmful.

- All commonly used medications to control asthma are safe in pregnancy.
- -All patients must be reassured that any flairs of their asthma must not be ignored and that treatment with medication such as steroids is safe both for themselves and for their fetus.
- -With regard to the effect of asthma on fetal outcome, there is no evidence that there is any significant impact on fetal growth or outcome.

- Any patient whose asthma seems to be deteriorating, particularly in the third trimester, should be seen by an obstetric physician for review.
- -It is obviously desirable that control of their asthma should be at its optimum prior to the onset of labour.

- **Patients presenting in labour should be managed:**
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 - 1.it is unusual for labour to be complicated by attacks of asthma and this is probably due to the increased secretion of cortisol during the process.
 - 2.However, attacks of asthma during labour can be managed by conventional treatment, such as inhaled betasympathomimetics.
 - -.

- Patients who have been on maintenance glucocorticoids, for example, Prednisolone doses in excess of the equivalent of 5 mg Prednisolone daily, require hydrocortisone cover during labour.
- If an operative delivery is required, epidural anaesthesia is preferable to general anaesthesia



Features of severe lifethreatening asthma

- Peak expiratory flow rate 4.6 kPa.
- Silent chest.
- Cyanosis.
- Bradycardia.
- Arrhythmia.
- Hypotension.
- Exhaustion.
- Confusion.

Tuberculosis

- -Pulmonary tuberculosis can present for the first time in pregnancy and the obstetrician must have a high index of suspicion when presented with symptoms of: cough, malaise or weight loss in high-risk groups.
- Most treatment options appear to be safe including: 1. ethambutol. 2.rifampicin. 3. isoniazid . 4. pyridoxine and also pyrazinamide.

TB

- Streptomycin carries risks of VIII nerve damage and should be avoided.
- -There is no conclusive evidence that the outcome of pregnancy is adversely affected by tuberculosis providing treatment is commenced in the first half of pregnancy.
- -After birth, the neonate should be treated with prophylactic isoniazid for 3 months and there after BCG vaccination should be given, although its efficacy remains questionable



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Neurology

Epilepsy

- Approximately 30% of those with epilepsy are women in their childbearing years, which means 1 in 200–250 pregnancies occur in women with a history of epilepsy.
- Pregnancy has no consistent effect on epilepsy: some women will have an increased frequency of fits, others a decrease and some no difference.
- Nonetheless, there is a 10-fold increase in mortality among pregnant women with epilepsy, and 1 in 20 indirect maternal deaths occur in women with epilepsy.

The principles of epilepsy management are that while the risks to pregnancy from seizures outweigh those from anticonvulsant medication, seizures should still be controlled with the minimum possible dose of the optimal drug.

The principal concern related to epilepsy in pregnancy is the increased risk of congenital abnormality caused by anticonvulsant medications.

- All of these drugs are associated with a 2–3-fold increased risk of fetal abnormality (5–6%) compared with unexposed epileptic mothers

- . Pregnant women with epilepsy who do not take antiepilepsy drugs (AEDs) do not expose their offspring to an increased risk of congenital abnormalities.
- Polytherapy increases the risk of major congenital abnormality by about 3% for each additional AED.
- The major fetal abnormalities associated with anticonvulsant drugs (including sodium valproate, carbamazepine, phenytoin, phenobarbitone) are neural tube defects, facial clefts and cardiac defects.
- .

- Many of these abnormalities are detectable by ultrasound and therefore all women should be offered detailed anomaly scanning

- In addition, each drug is associated with a specific syndrome that includes developmental delay, nail hypoplasia, growth restriction and midface abnormalities.
- In the case of valproate, the likelihood of these effects is dose dependent ($>1,000$ mg/day) and it should be avoided in pregnant women, except when epilepsy cannot be controlled with other AEDs

Causes of seizures in pregnancy

- Epilepsy.
- Eclampsia.
- Encephalitis or meningitis.
- Space-occupying lesions (e.g. tumour, tuberculoma).
- Cerebral vascular accident.
- Cerebral malaria or toxoplasmosis.
- Thrombotic thrombocytopaenic purpura.
- Drug and alcohol withdrawal.
- Toxic overdose.
- Metabolic abnormalities (e.g. hypoglycaemia).



Prepregnancy counselling in epilepsy

- Alter medication according to seizure frequency.
- Reduce to monotherapy where possible.
- Stress importance of compliance with medication.
- Preconceptional folic acid 5 mg.
- Explain risk of congenital malformation.
- Explain risk from recurrent seizures.